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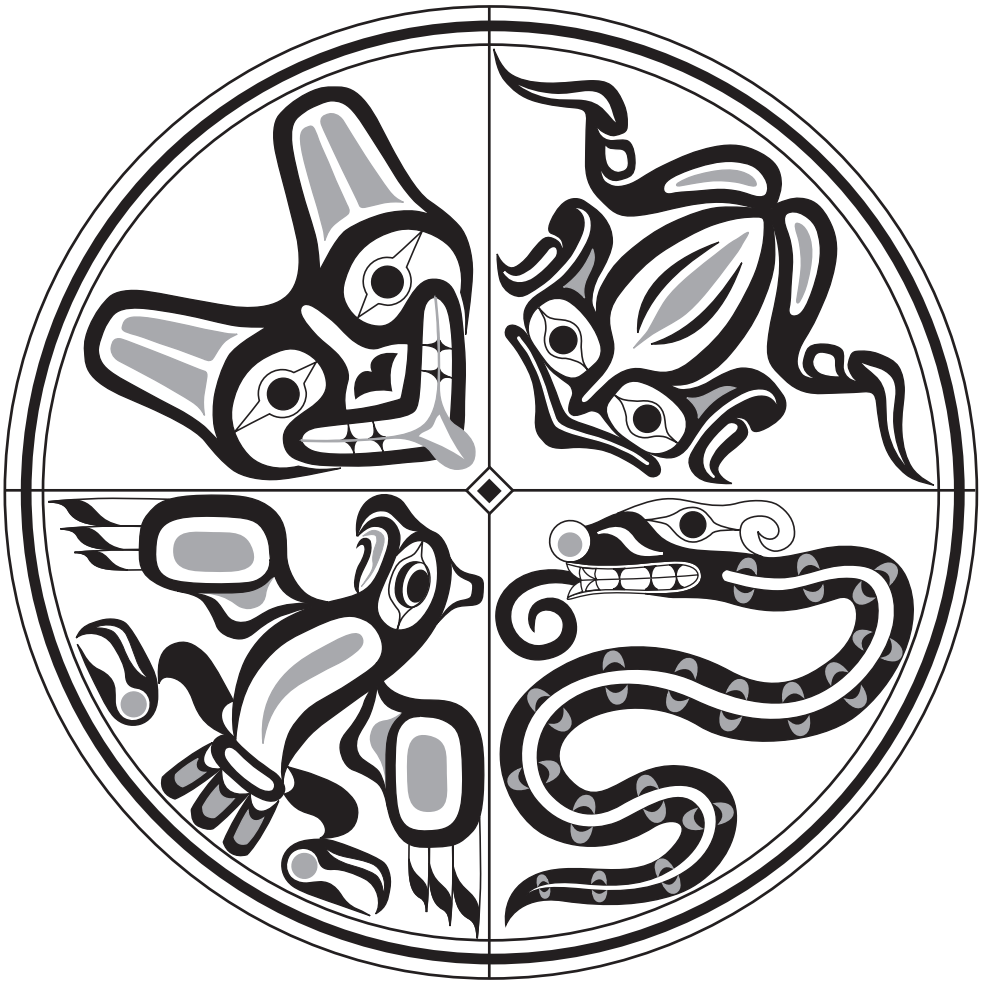

SUMMARY OF BIRDS KILLED BY A HARMFUL ALGAL BLOOM ALONG THE SOUTH WASHINGTON AND NORTH OREGON COASTS DURING OCTOBER 2009[1]

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Key words: *Akashiwo sanguinea*, demography, harmful algal bloom, Oregon, plumage fouling, seabird, Washington

Seabirds are known to be vulnerable to biotoxins produced by harmful algal blooms (HABs). The ingestion of prey contaminated with biotoxins such as domoic acid can cause disorientation, seizures, morbidity, and mortality (Work and others 1993). During November 2007 in Monterey Bay, California, an unprecedented stranding of live and dead seabirds highlighted plumage fouling as an emerging mortality factor during harmful algal bloom (HAB) events. The fouling agent was a proteinaceous foam derived from the cellular breakdown of the dinoflagellate *Akashiwo sanguinea*. This foam coated seabird plumage in a manner similar to fouling by petroleum oil (Hartung 1967; McEwan and Koelink 1973; National Research Council 1985), causing waterlogging, hypothermia, morbidity, and mortality in the California birds (Jessup and others 2009).

During late October 2009 along the southern Washington and northern Oregon State coasts, algal blooms caused another seabird plumage fouling event, where several bird species were coated in a foam surfactant. Water and feather samples confirmed that this dinoflagellate bloom was also *A. sanguinea* (Dr. Raphael Kudela, University of California Santa Cruz, Santa Cruz, CA, pers. comm.). Du and others (in press) describe the conditions that facilitated the *A. sanguinea* bloom off the Oregon coast and propose that the source of the bloom was southward transport of cells from a massive September bloom that first occurred off the Washington coast. Similar to the 2007 California event, the foam produced by the Washington-Oregon bloom reduced the waterproofing of birds' plumage, causing hypothermia in individual birds and leading to the stranding of hundreds of debilitated live and dying seabirds on public beaches.

During this event, volunteers and federal, state, and local resource management agencies collected as many birds as possible for rescue and rehabilitation from the Long Beach Peninsula, WA; Fort Stevens State Beach, OR; and Cannon Beach, OR. Most of the birds collected were initially delivered to the Wildlife Center of the North Coast (WCNC) rehabilitation facility in Astoria, OR, where volunteers observed that birds were cold, weak, stressed, and starving. Due to the magnitude of this event, the WCNC facility was quickly overwhelmed by hundreds of collected seabirds, which included Common Loons (*Gavia immer*), Common Murres (*Uria aalge*), Pacific Loons (*G. pacifica*), Red-throated Loons (*G. stellata*), Surf Scoters (*Melanitta perspicillata*), and Western and Clark's Grebes (*Aechmophorus occidentalis* and *A. clarkii*). Birds were subsequently transferred to facilities that were better equipped to handle large numbers of seabirds for rehabilitation, including the International Bird Rescue Research Center (IBRRC) in Cordelia, CA (n = 479 birds), and Progressive Animal Welfare Society (PAWS) in Lynnwood, WA (n = 125). Some birds responded well to supportive care and were eventually

[1] We dedicate this manuscript to the men and women at US Coast Guard Air Station Sacramento, who donated their time and use of the C-130 Hercules aircraft to transport seabirds. This same Coast Guard aircraft crashed off the coast of San Diego, CA on 30 October 2009, within days of responding to the HAB event documented in this manuscript. All hands on board that aircraft were lost. We honor their memory and service.

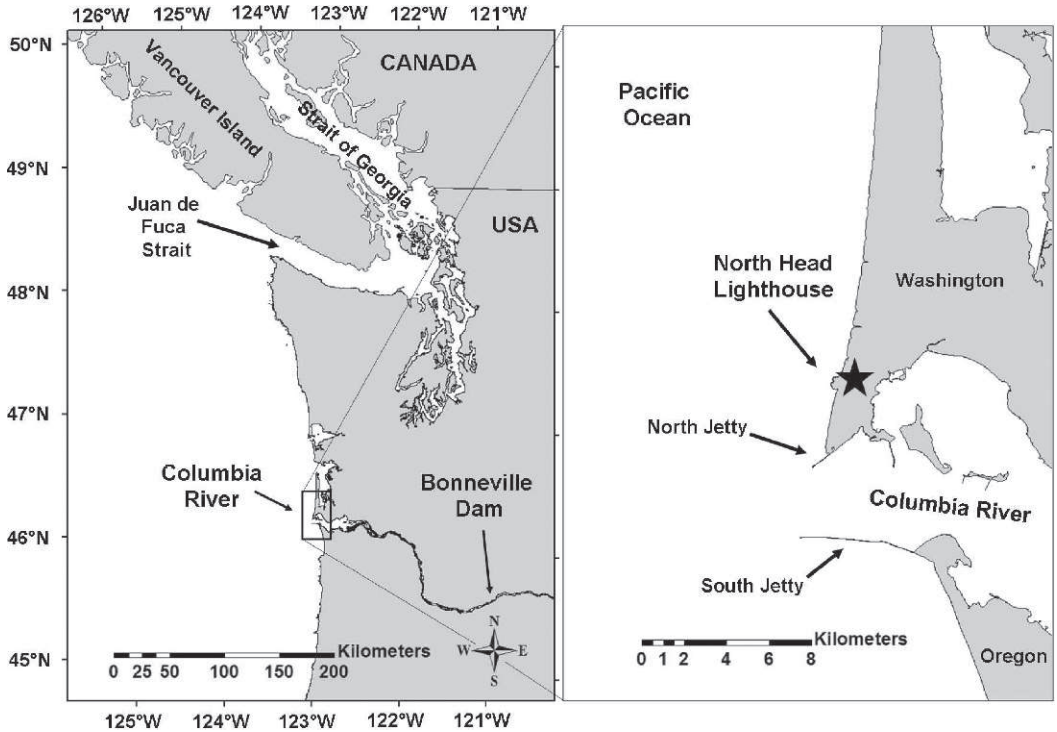

FIGURE 1. Map of coastal US and Canada and panel showing the observation site at North Head Lighthouse in Washington State.

released. However, many birds were too physiologically compromised and did not survive.

We summarize information from necropsies of birds that were collected dead from beaches or that died during rehabilitation in the Oregon (WCNC) and California (IBRRC) facilities. A total of 308 seabirds were examined at 2 facilities: NOAA Fisheries Point Adams Research Station, Hammond, OR (n = 100; NOAA) and California Department of Fish and Game Marine Wildlife Veterinary Care and Research Center, Santa Cruz, CA (n = 208; MWVCRC). We identified species of all carcasses and recorded information on morphometrics, age, sex, and body condition. Tissue samples from skeletal muscle, feathers, liver and in some cases, stomachs, were collected and stored frozen at MWVCRC.

Species composition of the examined birds was compared with data from local monthly marine bird surveys, which have been conducted since 2004 at the North Head Lighthouse in Washington State (UTM: Zone 10, 417191.55E, 5127731.94N, NAD1983; elevation 59 m above sea level). The survey site is located at the southern end of the Long Beach Peninsula, within the area where large numbers of birds stranded during the HAB event (Fig. 1). On each survey day, birds sitting on the water were counted every half hour between sunrise and 15:30 in the same 1.9-km^2 nearshore sector from the surf to 1.4 km offshore of the lighthouse.

To determine sex and approximate age of the seabirds killed during this event, we identified and assessed gonad maturity (length and width of gonad in both sexes, diameter of largest follicle and oviduct development in females) and recorded the presence or absence of the bursa of Fabricius, a lymphatic gland that is largest in 1st-year birds and atrophies with age and sexual maturation (Broughton 1994). When present, bursal length and width were recorded. We also noted plumage type (basic, transitional, alternate) and molt of the primaries and body feathers. We quantified body condition by measuring body mass as well as assigning each bird a numerical condition score based upon examination of body fat and muscle tissue.

TABLE 1. Total number (n), sex ratio, and estimated age of seabirds (listed alphabetically) examined at NOAA and MWVCRC following Washington–Oregon harmful algal bloom event in October 2009.

			Age Estimate		
Species	n	Sex (M:F ratio)	Adult	Immature	Unknown
Clark's Grebe	3	0.5:1	2	1	-
Common Loon	16	0.4:1	14	2	-
Common Murre	82	3.8:1	56	21	5
Pacific Loon	3	0.5:1	2	-	1
Red-Throated Loon	116	1.1:1	93	12	11
Surf Scoter	2	1.0:1	1	-	1
Western Grebe	86	0.8:1	42	36	8
Total	308		210	72	26

Because birds in deteriorating physiological condition metabolize fat reserves before metabolizing muscle proteins, this body condition score measures the relative availability of fat and muscle reserves (van Franeker 2004). We scored subcutaneous and intestinal fat between zero (poor condition; no fat) to three (good condition; very fat). When present, we measured subcutaneous fat depth in millimeters over the breast muscle to the right of the keel. Similarly, the condition of the breast muscle (pectoralis-supracoracoideus complex) was scored between zero (poor condition; strongly emaciated, muscle significantly atrophied below keel-line) to three (good condition; muscle at or above keel-line). Following van Franeker (2004), we then summed the three condition scores into an overall body condition index. During the examination we also noted any external injuries to wings, bill, or feet.

Red-throated Loons, Western Grebes, and Common Murres represented the greatest proportion of examined birds (92%, n = 284, Table 1). The majority of birds examined were adults (68%, n = 210), and 23% were classified as immature (n = 72). Eight percent (n = 26) were classified as unknown age, but were most likely immature or sub-adult birds. With the exception of Red-throated Loons, the species composition of birds examined was proportionally similar to that typically observed in local nearshore marine bird surveys (Fig. 2). Western Grebes are the dominant species observed in fall and winter, but loons, murres, and scoters are also regularly observed.

There were no apparent differences in the sex ratios of any of the seabird species examined, with the exception of Common Murres. Nearly 4 times more male than female murres were affected by this event (Table 1), and 68% were adults. Twenty-six percent of the murres examined were immature birds, and most of these were hatch-year chicks. The bias in murre sex ratio is explained by their breeding ecology: adult males accompany newly fledged chicks to sea and associate with the chick throughout the fall and winter season before leaving the chick to forage on its own (Scott 1990). Similar male-biased ratios in murres have been observed in other fall mortality events, including gill-net fisheries bycatch in California (Nevins 2004).

All examined birds were moderately to severely emaciated, with a high frequency of pectoral muscle atrophy and virtually no subcutaneous or internal fat (Table 2). On average, all birds examined were 31% below normal body mass, indicative of starvation stress. Although body condition indices improved with duration of rehabilitation, and the range of values for overall body condition index was 0 to 7, the median value for all species was within the critically to mortally emaciated range (0 to 2). With the exception of birds that were force-fed fish or given nutritional tube feedings while in rehabilitation, most birds had empty stomachs or only stones in their stomachs. Numerous Western Grebes had moderate to significant dry feather balls present in their ventriculi, a common observation in grebes, which are known to eat their feathers (Ehrlich and others 1988). In healthy grebes these feather balls are moist and saturated with digesta. Many of the birds examined had generalized pallor of muscle tissue and organs, as well as evidence of urate stasis in the kidneys, ureters, and cloacae, suggestive of dehydration. In particular, 53% of Red-throated Loons had distended cloacae, perhaps related to the build

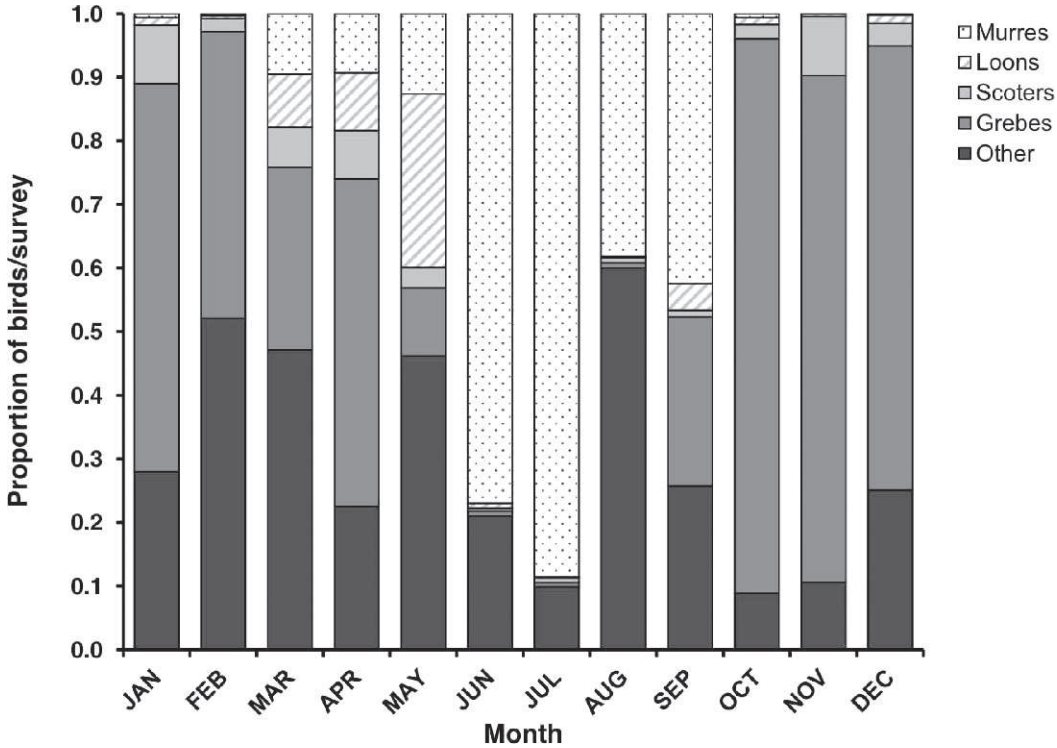

FIGURE 2. General taxonomic composition of seabirds observed from the North Head Lighthouse in Washington State. Proportions were calculated from mean maximum daily counts of sitting birds for each month from August 2004 to December 2009.

up of crystallized urates. Similarly, 50% of Western Grebes and 16% of Common Murres had distended cloacae. Additionally, we commonly found gastrointestinal hemorrhage, another indicator of physiological stress.

Grebes, murres, and loons are essentially flightless during their primary molt, which occurs soon after their arrival in wintering habitat along the coasts of Washington and Oregon. This makes them more susceptible to oiling or plumage fouling events due to lack of mobility. Molting birds are also likely under greater nutritional stress than non-molting birds. Fifty-eight percent of the birds were

TABLE 2. Summary of body condition measures for all seabird species (listed alphabetically) examined at NOAA and MWVCRC following Washington–Oregon harmful algal bloom event in October 2009.

Species	Mean body mass (g)	Percent below normal mass[a]	Mean fat depth (mm)	Median muscle score (0–3)	Median subcutaneous fat score (0–3)	Median intestinal fat score (0–3)	Median overall body condition (0–9)
Clark's Grebe	731.7	47.7	0.0	1	0	0	1.0
Common Loon	2673.4	34.8	0.2	1	0	0	1.5
Common Murre	657.3	33.6	0.0	0	0	0	0.0
Pacific Loon	1358.3	20.1	0.0	0	0	0	0.0
Red-throated Loon	1249.0	10.8	0.0	0	0	0	0.0
Surf Scoter	700.0	26.3	0.0	0	0	0	0.0
Western Grebe	847.2	43.5	0.0	0	0	0	0.0

[a] Normal body mass is based on Sibley (2000).

undergoing molt of the primary feathers, and almost all of the birds examined were in winter or basic plumage. Because many of these birds were actively undergoing molt, sensitive blood feathers were damaged during the stranding event. Injuries to primary feathers and wings were documented in the three most common species: 24% of Red-throated Loons had moderate to severe injuries to their wings; 9% of Western Grebes and 4% of Common Murres also had wing injuries. These injuries most likely occurred during beach stranding or during active avoidance of humans or predators while stranded. It is also possible some injuries occurred during capture and handling. Additional wing injuries noted during the necropsies included broken or damaged primaries and abrasions along the carpal joints.

Reports from rescue organizations indicated that Red-throated Loons and Western Grebes were the most difficult to capture, and were therefore the most likely to be left on beaches. Many loons travelled on land into residential and business neighborhoods, up to a half-mile from the beach, where cars struck some of the birds. Loons and grebes are fully aquatic species that never come ashore, and their bone structure is not designed to support their body weight on land. Therefore, these species are particularly vulnerable to injury caused by abnormal weight bearing on leg and keel bones. Most foot and keel injuries were likely attributable to bird movement from sea to town (Sharnelle Fee, Wildlife Center of the North Coast, Astoria, OR, pers. comm.). The primary reason birds in rehabilitation were euthanized was due to the severity of leg or keel injuries. Fifty-six percent of Red-throated Loons necropsied at MWVCRC had severely edematous or necrotic foot webbing and 45% had four or more fractured or luxated terminal toe joints. Thirty-one percent of loons had significant keel pressure sores. Additionally, 13% of Common Loons and 5% of Red-throated Loons and Western Grebes had moderate to severe injuries to their bills, which included broken bill tips and fractured mandibles. Bill injuries also likely occurred during stranding, especially if the birds were weakened and traveling over hardened surfaces. Proper handling of birds, including the use of containers with narrow bar spacing to prevent birds from sticking long bills or necks through the bars, should reduce the likelihood of bill injury during capture and transport.

The similarity of species composition between local survey data and the HAB event suggests that impacts of this mortality event were nonselective in the nearshore environment. The higher proportion of Red-throated Loons affected by the HAB is most likely explained by the temporal and spatial extent of the HAB event coinciding with a pulse of migrating loons. The seabirds affected are also those typically affected by winter oil spills, including nearshore species such as loons and grebes (Hampton and others 2003). Jessup and others (2009) noted that most foam observed during the 2007 California event occurred in the surf zone, where turbulent mixing of seawater by wave action likely created the foam. However, there may be unobserved impacts on other species that did not strand on the beaches, including seabirds that typically occur further offshore and may not be regularly detected in nearshore or beach surveys. For example, Jessup and others (2009) noted that pelagic species including the Northern Fulmar (*Fulmarus glacialis*) seemed to have encountered the foam further offshore, where foam accumulated along frontal edges of the bloom.

Similar HAB events may be increasing in frequency (Parsons and others 2006; Jessup and others 2009). Because such events could be linked to regional climate change (Cloern and others 2005; Moore and others 2009), coastal development, or changes in nearshore water chemistry (Paerl 1997; Sellner and others 2003), it is important to accurately document the timing, geographic extent, duration, and ecosystem impacts. Past studies have found that many HAB-associated seabird mortality events are underreported (Shumway and others 2003). Thus, a rapid, coordinated response by local and regional wildlife rehabilitation groups, biologists, and natural resources agencies is required, and strandings of particularly sensitive or threatened species are important to document. For example, the preponderance of mature birds affected by the 2009 event may be particularly important for Red-throated Loons, which face a multitude of conservation threats on breeding grounds (Barr and others 2000). In conjunction with rescue efforts during a HAB event, collecting data on the species composi-

tion and estimated number of birds affected, as well as the spatio-temporal impact of the event, will be required if natural resource agencies are to determine the demographic effects of such events on marine bird populations. This latter type of scientific data collection typically falls outside the normal focus and capabilities of wildlife rescue or rehabilitation facilities.

The impacts of HAB events on seabird populations are not well known, and if the frequency or magnitude of HAB events is increasing, HABs could represent another mortality source of conservation concern. Most of the species affected by this event face other conservation threats including oil spills, habitat degradation, pollution, fisheries interactions, and introduced predators at their colonies. Although response networks are in place for documenting anthropogenic events such as oil spills, there are not analogous response networks in place for documenting unusual die-off events, including HABs. We recommend that agencies build upon successful oil-spill response network models and provide support for developing networks and establishing common data collection protocols. In the meantime, we encourage continuing regional cooperation to document any similar events which may occur in the future.

Acknowledgments.—We thank the many staff and volunteers at regional rehabilitation centers for their efforts to collect, transport, and rehabilitate birds and to provide specimens for necropsy, including: the Wildlife Center of the North Coast (S Fee), the International Bird Rescue Research Center (J Holcomb), the Oiled Wildlife Care Network (M Ziccardi) and the members of the US Coast Guard who assisted with transporting birds (Districts 11 and 13). D Jacques (Pacific Eco Logic) responded rapidly to collect specimens for necropsy during this unusual mortality event; we are also grateful for her assistance during necropsies at NOAA. We thank all volunteers who helped with necropsies: S Hinton and M Morrow at NOAA, and F Batac, E Donnelly, B Hoover, A Ford, A Wells, and C Young at MWVCRC. Necropsy work and report compilation were supported by funding from NOAA Northwest Fisheries Science Center, Office of the Science Director (to J Zamon), Moss Landing Marine Laboratories (to J Harvey), and the California Department of Fish and Game, Marine Wildlife Veterinary Care and Research Center (to D Jessup). W Peterson and 2 anonymous reviewers provided helpful comments on this manuscript.

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Cooperative Institute for Marine Resources Studies, Oregon State University, Hatfield Marine Science Center, 2030 Marine Science Drive, Newport, OR 97365 (EMP); Point Adams Research Station, NOAA Northwest Fisheries Science Center, 520 Heceta Place, Hammond, OR 97121; jen.zamon@noaa.gov (JEZ); Marine Wildlife Veterinary Care and Research Center, California Department of Fish and Game, 1451 Shaffer Road, Santa Cruz, CA 95060 (HMN, CMG); International Bird Rescue Research Center, 4369 Cordelia Road, Fairfield, CA 94534 (RSD); Pacific States Marine Fisheries Commission, 205 SE Spokane Street, Portland, OR 97202 (LHK). *Submitted 13 October 2010, accepted 13 December 2010. Corresponding Editor: Joan Hagar.*

ERRATUM:

The following is a correction to a GENERAL NOTE published in the Autumn 2011 (Volume 92 Number 2) issue titled SUMMARY OF BIRDS KILLED BY A HARMFUL ALGAL BLOOM ALONG THE SOUTH WASHINGTON AND NORTH OREGON COASTS DURING OCTOBER 2009 by Elizabeth M Phillips, Jeanette E Zamon, Hannahrose M Nevins, Corinne M Gibble, Rebecca S Duerr, and Lauren H Kerr. Paragraph 3 on page 120 (right column), beginning with the 4th sentence, should have been corrected to:

''Of the 792 total birds collected during this event, 597 were subsequently transferred to facilities that were better equipped to handle large numbers of seabirds for rehabilitation, including the International Bird Rescue Research Center (IBRRC) in Cordelia, CA (n = 479 birds) and Progressive Animal Welfare Society (PAWS) in Lynnwood, WA (n = 118). Some of the collected birds responded well to supportive care and were eventually released (n = 344). Many birds (n = 448), however, were too physiologically compromised and did not survive.''